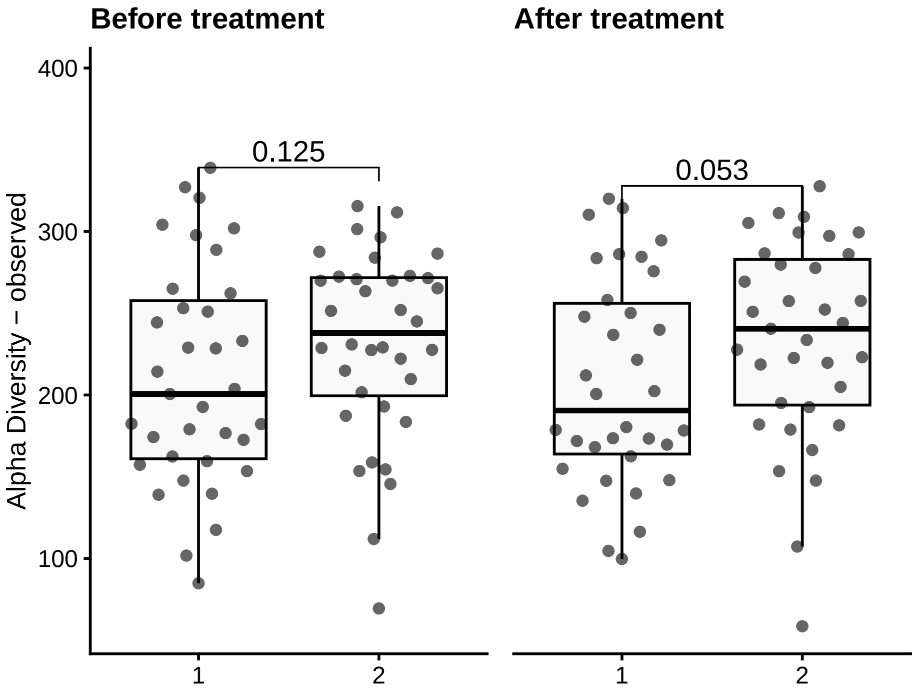
Alpha Diversity

Muluh

1 Community diversity

2 Group-wise comparisons

- **Index:** observed
- **Group variable:** diet
- **Multiple testing correction:** fdr
- **Statistical test** Wilcoxon



2.1 Table of the top findings

2.1.1 Across-group

2.1.1.1 Before treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

group1	group2	mean1	mean2	log2FC	p.adj
1	2	211.1	231.6	0.13	0.125

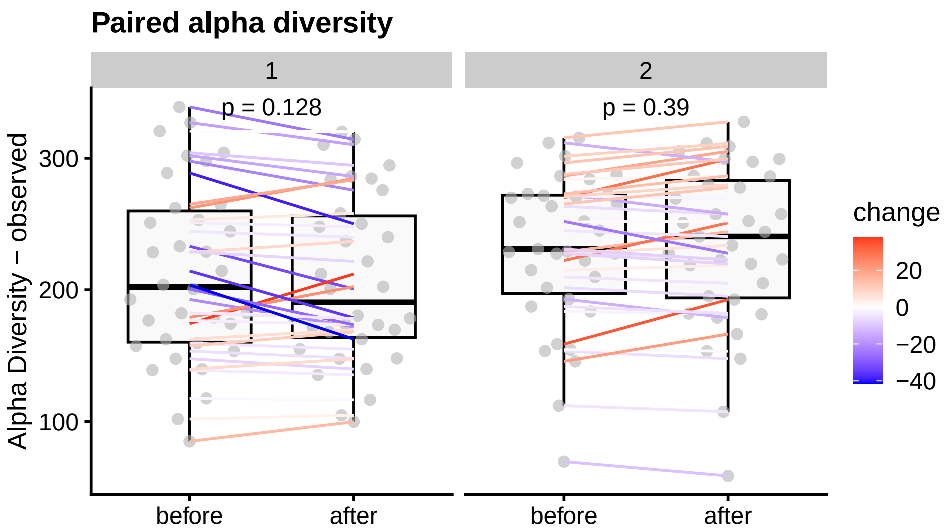
2.1.1.2 After treatment

- Multiple testing correction: **fdr**

- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
1	2	207.1	233.3	0.17	0.053



2.1.2 Within-group

2.1.2.1 Paired treatment

- Multiple testing correction: **paired** -none
- Statistical test: **Wilcoxon**

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

diet	mean1	mean2	log2FC	p.value
1	211.1	207.1	0.04	0.128
2	231.6	233.3	0.13	0.390

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: y ~ diet * time + (1 | id)
   Data: df_no_miss

REML criterion at convergence: 7.7

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.88441 -0.35860  0.00522  0.37488  1.86733

Random effects:
 Groups   Name        Variance Std.Dev.
 id       (Intercept) 0.215777 0.46452
 Residual             0.006819 0.08258
Number of obs: 140, groups:  id, 71

Fixed effects:
            Estimate Std. Error       df t value Pr(>|t|)
(Intercept)  7.67293    0.08456 88.10071  90.739   <2e-16 ***
```

Characteristic	before			after		
	Beta	95% CI[1]	p-value	Beta	95% CI[1]	p-value
(Intercept)	151	67, 236	<0.001	144	59, 229	0.001
meal_group			0.4			0.3
1	—	—		—	—	
2	32	-11, 75	0.14	28	-16, 72	0.2
3	28	-15, 71	0.2	30	-14, 74	0.2
4	10	-32, 53	0.6	-3.9	-47, 39	0.9
age	1.0	-0.44, 2.5	0.2	1.2	-0.25, 2.7	0.10

[1]CI = Confidence Interval

Paired test between timepoints (1 and 2)	Beta	95% CI[1]	p-value
(Intercept)	147	84, 210	<0.001
meal_group			0.6
1	—	—	
2	28	-16, 71	0.2
3	28	-15, 71	0.2
4	2.1	-28, 32	0.9
age	1.2	0.13, 2.2	0.025

[1]CI = Confidence Interval

```
diet2        0.11530    0.11875 88.09749   0.971    0.334
time        -0.02737    0.02002 67.10134  -1.367    0.176
diet2:time   0.03662    0.02811 67.10054   1.302    0.197
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
           (Intr) diet2  time
diet2      -0.712
time       -0.352  0.251
diet2:time  0.251 -0.352 -0.712
```

3 Alternate analyses

- **Beta (Estimate)**: Shows the effect size and direction of a predictor on the outcome. Positive means an increase, negative a decrease.
- **95% CI**: Range where the true beta is likely to fall, reflecting precision.
- **p-Value**: Tests if beta differs from zero; p<0.05p < 0.05p<0.05 suggests significance.

4 Data Summary

```
class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
```

```
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(15): sample id ... group observed_log2
reducedDimNames(0):
mainExpName: NULL
altExpNames(14): kingdom phylum ... KO metacyc
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```